POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Applied Physics | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is simple harmonic motion? Derive an expression for the time period of a bar pendulum and show that the point of suspension and oscillation are interchangeable. 9

OR

What is torsional pendulum? Derive an expression for time period of torsional pendulum. Find the expression for modulus of rigidity of the wire.

b) Find the expression for experimental value of g (acceleration due to gravity) by using time period of compound pendulum. 6

2. a) What is reverberation time? Find the expression for Sabine's Formula. What are it's significance in our daily life? 9

b) The size of an empty assembly of bell has dimensions 20X15X5 cm^3 and the reverberation time is 3.5 sec. What area of the wall should be covered by curtain cloth to reduce the reverberation time by 2.5 sec; if the absorption coefficient of the curtain cloth is 0.5. Also calculate the average absorption coefficient of the bell. 6

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3. a) What is LASER? Explain the construction & working principle of He-Ne Laser with necessary energy level diagram. Also write the applications of LASER. 9

b) What do you mean by acceptance angle & numerical aperture in optical fibre. An optical fiber has fractional refractive index change of 0.2 and a cladding refractive index 1.59. Determine acceptance angle for the fiber in water, which has refractive index of 1.33. Also evaluate Numerical Aperture (N.A.). 6

4. a) What is Capacitor? Discuss charging and discharging of capacitors. 9

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b) Obtain the time constant of a capacitor in RC circuit such that the current through the resistor increases to one third of its peak value in 5 seconds. 6

5. a) What is Maxewell Equation? Write Maxwell's equations in integral and differential forms? Convert the integral form of Maxwell's equation into it's differential form. 9

b) Using Maxwell's equation, Prove that $C = \frac{1}{\sqrt{\mu_o \epsilon_o}}$ Where symbols carry their usual meaning. 6

6. a) What is thermodynamic process? Differentiate between reversible and irreversible process. Find the expression for coefficient of performance of Refrigerator and Heat Engine. 9

b) A brick wall 12 cm thick and 5 m^2 surface area is exposed to 250^{0}C at one face and 50 ^{0}C to another face. If the thermal Conductivity of the material is 1.5 W/mK, Determine the heat transfer rate. 6

OR

An ideal gas at 17^{0}C has a pressure of 760 mm of Hg is compressed adiabatically until it's volume is halved. Calculate the pressure and temperature of gas.

2×5

7. Write short notes on: **(Any two)**
   a) Time Independent SWE
   b) Progressive wave
   c) Clausius-Mossotti equation

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